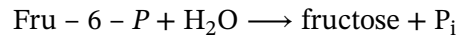


Fructose 6-phosphate (Fru-6-*P*) breaks down in water (undergoes hydrolysis) according to the reaction below.



The equilibrium constant and the Gibbs free energy of the reaction are given by Equations 1 and 2, respectively.

$$K_{\text{eq}} = \frac{[\text{fructose}][\text{P}_i]}{[\text{Fru} - 6 - P]}$$

Equation 1

$$\Delta G^\circ = -15 \text{ kJ/mol}$$

Equation 2

If $[\text{Fru-6-P}] = x \text{ mol/L}$ and $[\text{P}_i] = 1.0 \text{ mol/L}$ at equilibrium, which expression gives the equilibrium concentration of fructose? (Note: Use $RT = 2.5 \text{ kJ/mol}$.)

- ☐ A. $(1/x)e^{-6}$ [16%]
☐ B. xe^{-6} [22%]
☐ C. $(1/x)e^6$ [16%]
☒ D. xe^6 [44%]

Correct

44% answered correctly

Explanation:

Equilibrium concentrations for the hydrolysis of fructose 6-phosphate

The diagram illustrates the equilibrium reaction: $\text{Fru-6-P} + \text{H}_2\text{O} \xrightleftharpoons{K_{\text{eq}}} \text{Fructose} + \text{P}_i$. It includes the equilibrium constant expression $K_{\text{eq}} = \frac{[\text{fructose}][\text{P}_i]}{[\text{Fru-6-P}]}$ and the relationship $\Delta G^\circ = -RT \ln K_{\text{eq}}$. The derivation shows the substitution of $\Delta G^\circ = -15 \text{ kJ/mol}$ and $RT = 2.5 \text{ kJ/mol}$ into the equation, leading to $6.0 = \ln \left(\frac{[\text{fructose}]}{x} \right)$, which simplifies to $e^6 = \frac{[\text{fructose}]}{x}$ and finally $xe^6 = [\text{fructose}]$.

The **equilibrium constant** K_{eq} for a reaction is expressed as a **ratio** of the **molar concentrations** of the products over the reactants, each raised to the power of its respective balanced reaction **coefficient** (in a hydrolysis reaction, water is the solvent and is omitted). The K_{eq} is related to the **standard Gibbs free energy** ΔG° of the reaction by the relationship:

$$\Delta G^\circ = -RT \ln K_{\text{eq}}$$

where R is the **gas constant** and T is the **Kelvin temperature**. Solving this equation for K_{eq} gives:

$$-\frac{\Delta G^\circ}{RT} = \ln K_{\text{eq}}$$

$$e^{-\frac{\Delta G^\circ}{RT}} = K_{\text{eq}}$$

Substituting the equivalent concentration ratio for the K_{eq} of the hydrolysis of fructose 6-phosphate (Fru-6-P) into the equation relates the concentrations of the chemical species to ΔG° as:

$$\frac{[\text{fructose}][P_i]}{[\text{Fru} - 6 - P]} = e^{-\frac{\Delta G^\circ}{RT}}$$

Finally, solving for [fructose] and substituting the given values for [Fru-6-P], $[P_i]$, and RT into the equation yields:

$$[\text{fructose}] = \frac{[\text{Fru} - 6 - P]}{[P_i]} \cdot e^{-\frac{\Delta G^\circ}{RT}}$$

$$[\text{fructose}] = \frac{x \text{ mol/L}}{1.0 \text{ mol/L}} \cdot e^{\frac{-(-15 \text{ kJ/mol})}{2.5 \text{ kJ/mol}}} = \frac{x}{1.0} \cdot e^{\frac{+15}{2.5}} = xe^6$$

Therefore, the expression for the molar concentration of fructose at equilibrium is xe^6 .

(Choice A) Both the leading coefficient and the sign of the exponent are incorrect. These errors result from using an inverted K_{eq} ratio in the initial calculation setup and from algebra errors that give an inverted [Fru-6-P]/ $[P_i]$ ratio when solving for [fructose] in the last step of the calculation.

(Choice B) The sign of the exponent is incorrect. This error can be obtained from failing to cancel the negative signs in the $-\Delta G^\circ$ term during the calculation or from using an **inverted K_{eq}** ratio in the initial substitution.

(Choice C) The leading coefficient of the expression is incorrect. This error could result from algebra errors that give an inverted [Fru-6-P]/ $[P_i]$ ratio when solving for [fructose].

Educational objective:

The equilibrium constant K_{eq} for a reaction is expressed as a ratio of the molar concentrations of the products over the reactants, each raised to the power of its respective balanced reaction coefficients. The standard Gibbs free energy ΔG° and the K_{eq} are related via the gas constant R and Kelvin temperature T by the equation $\Delta G^\circ = -RT \ln K_{eq}$.

Time spent: 0 sec

Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws